

6	(a)	(i)	Area under graph from 0 to 5.0 m $= \frac{1}{2} (5) (200 + 160)$ $= 900 \text{ J}$ (correct substitution + final answer)	1
		(ii)	Gain in KE of child = Net work done $= \text{Work done by driving force} - \text{work done by friction}$ $\frac{1}{2} m v^2 = 900 - (160 \times 5)$ $v = \sqrt{100 (2) / 30.5} = 2.56 \text{ m s}^{-1}$ (correct substitution + final answer) Alternative: $KE_i + E_{\text{supplied}} = KE_f + E_{\text{dissipated}}$	1 1
	(b)		$(2m) c^2 = 3.38 \times 10^{-11}$ (correct expression) $m = 1.88 \times 10^{-28} \text{ kg}$ (correct final answer, 3 s.f.)	1 1
	(c)		${}_0^1\text{n} + {}_1^2\text{H} \rightarrow {}_1^3\text{H}$ (accept ${}_1^3\text{X}$) ${}_1^3\text{H} \rightarrow {}_{-1}^0\text{e} + {}_2^3\text{He}$ (do not accept ${}_2^3\text{X}$) Final isotope : ${}_2^3\text{He}$ or He-3	1 1

	(d)	Difference in mass between parent and products is fixed. Thus, <u>the corresponding energy released, which is mainly in the form of KE of the beta particle</u> should also be a constant. The continuous spread of energy values seemed to <u>violate the principle of conservation of energy</u>.	1 1
	(e)	(i) $\frac{N_K}{N_{Ar}} = \frac{1}{6} \Rightarrow \frac{N_K}{N_{Ar} + N_K} = \frac{1}{7}$ $\Rightarrow \frac{N_K}{N_{K_0}} = \frac{1}{7}$ $\Rightarrow e^{-0.693/(1.37 \times 10^9) t} = \frac{1}{7} \quad (\text{correct expression, substitution})$ $\Rightarrow t = 3.85 \times 10^9 \text{ years} \quad (\text{correct final answer, 3 s.f.})$	1 1
		(ii) If the escaped argon was accounted for, the ratio will be smaller than 1/6. Thus time elapsed is actually greater than 3.85×10^9 years. Thus, the answer in (e)(i) is an underestimate.	1 1